ARMY 2040: WORLD-CLASS. MULTI-MISSION READY. CROSS-DOMAIN CAPABLE.

# ELECTRONIC INTELLIGENCE (ELINT)

*Reading the enemy's radars from the ground*

**SOAC**

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03 Nov 2026 · SIGNAL SCHOOL, ITG, ETC, PA

HONOR. PATRIOTISM. DUTY.

# Lesson Overview and Objectives

**TERMINAL LEARNING OBJECTIVE**

Upon completion, students will explain the purpose of ELINT, describe its basic collection and analysis process, and identify the radar characteristics used to recognise a threat.

## By the end of this lecture you can:

**ELO 1**

Define ELINT and its place within SIGINT and EW

**ELO 2**

Distinguish ELINT from COMINT, and ES from ELINT

**ELO 3**

Describe how ELINT is collected, analysed and reported

**ELO 4**

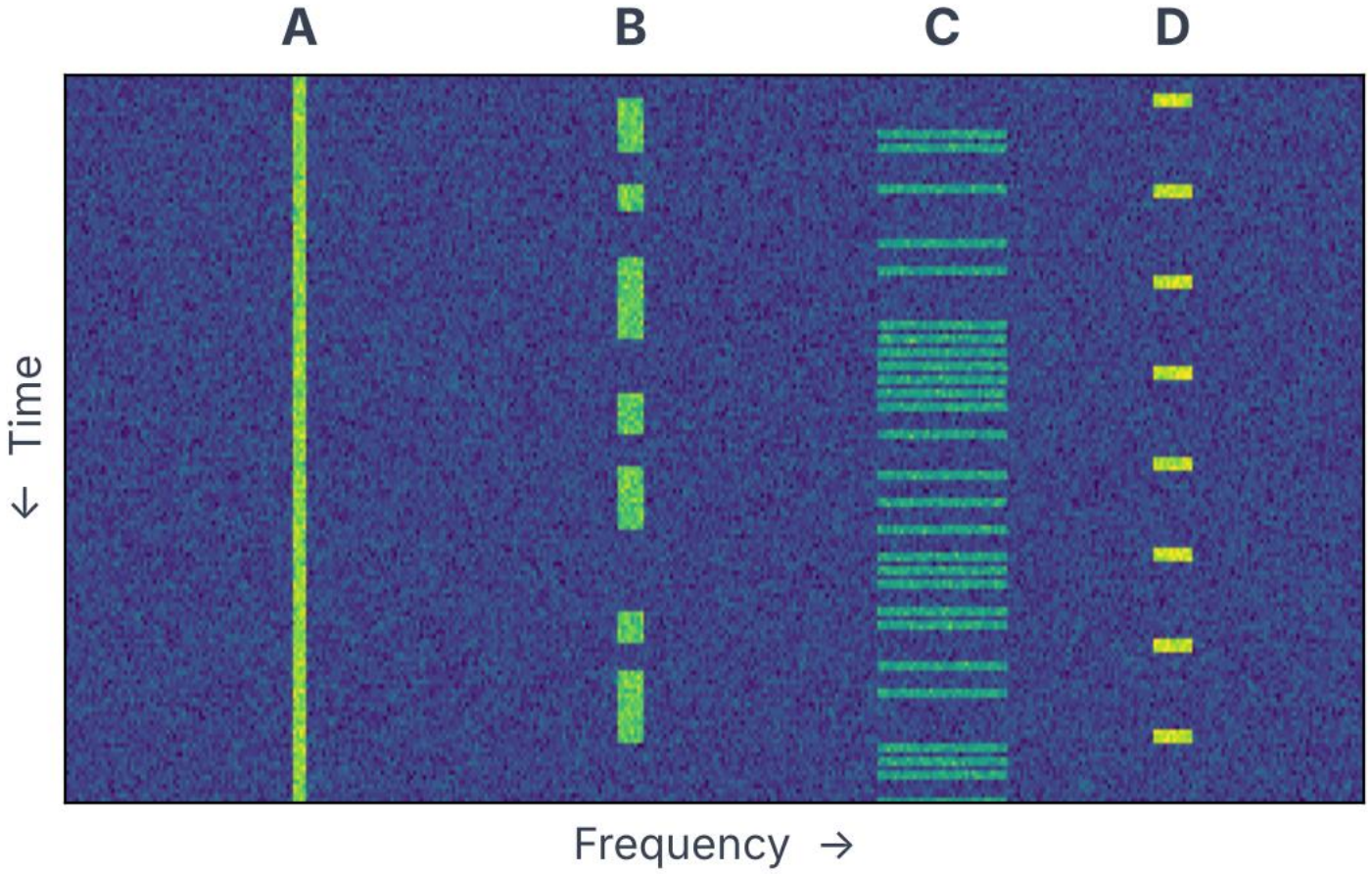
Identify radar parameters and what they reveal

**DOCTRINAL BASIS**

**PAM 3-0702-01 Electronic Warfare Operations (Sep 2026)**

Phones on silent · Questions anytime · About 45 min

# Ice Breaker: Find the Radar



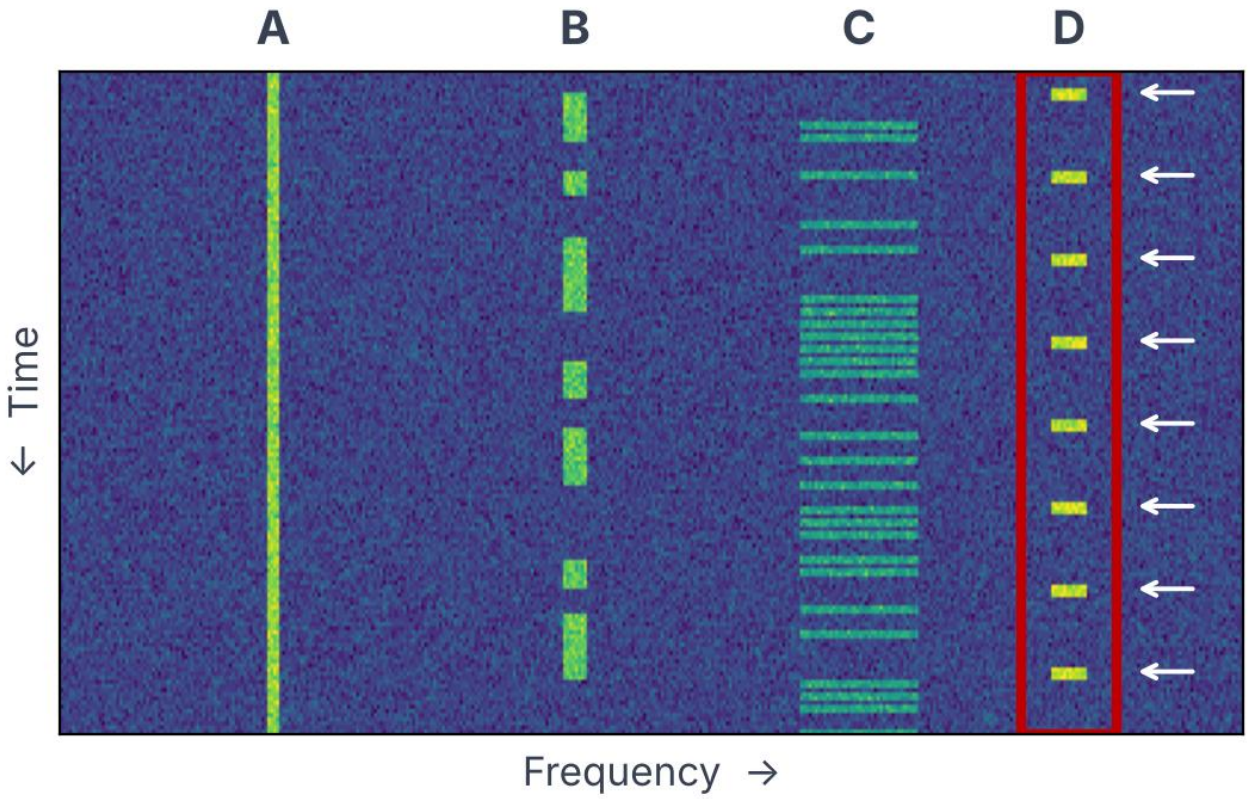

*Waterfall: newest at the top; bright = strong signal. Teaching illustration.*

**ICE BREAKER**

**This waterfall shows four emitters over time. Which one is most likely a rotating radar, and why?**

- **A** Steady line
- **B** Irregular bursts
- **C** Many short wideband bursts
- **D** Short, evenly spaced hits

# Ice Breaker: The Answer

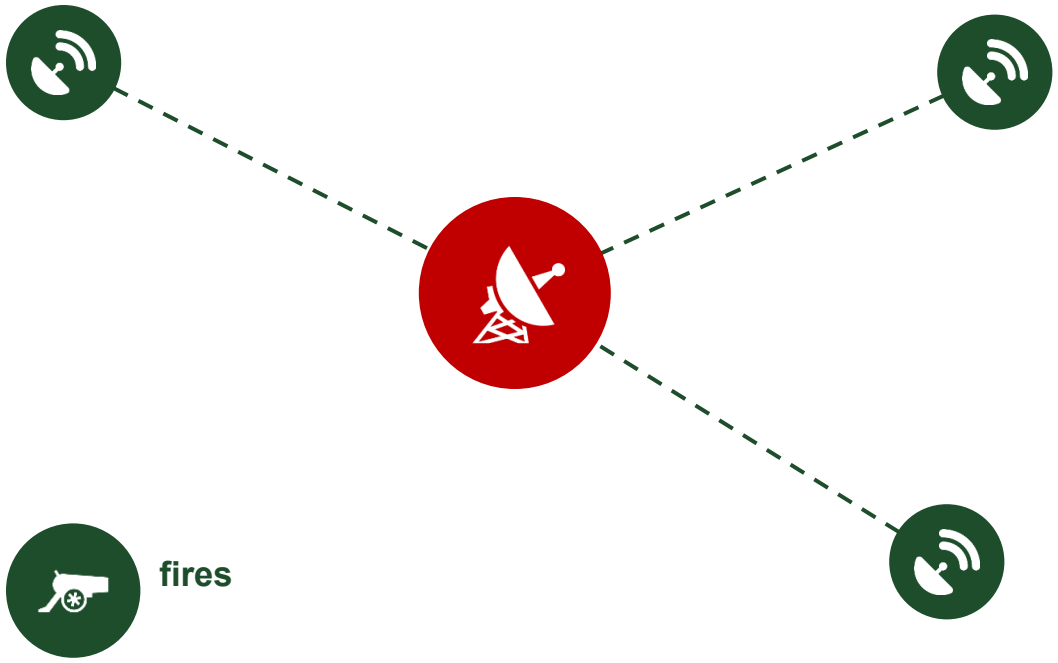



**D. The hits repeat at exact intervals: a beam sweeping past us once per rotation.**

- **A** Steady carrier: a beacon or a radio left on
- **B** Irregular bursts: someone talking (push-to-talk)
- **C** Bursty wideband: data traffic
- **D** **Machine rhythm: a rotating radar**

**People talk irregularly. Machines repeat. ELINT reads the machine's rhythm.**

# Why ELINT Matters on the Ground



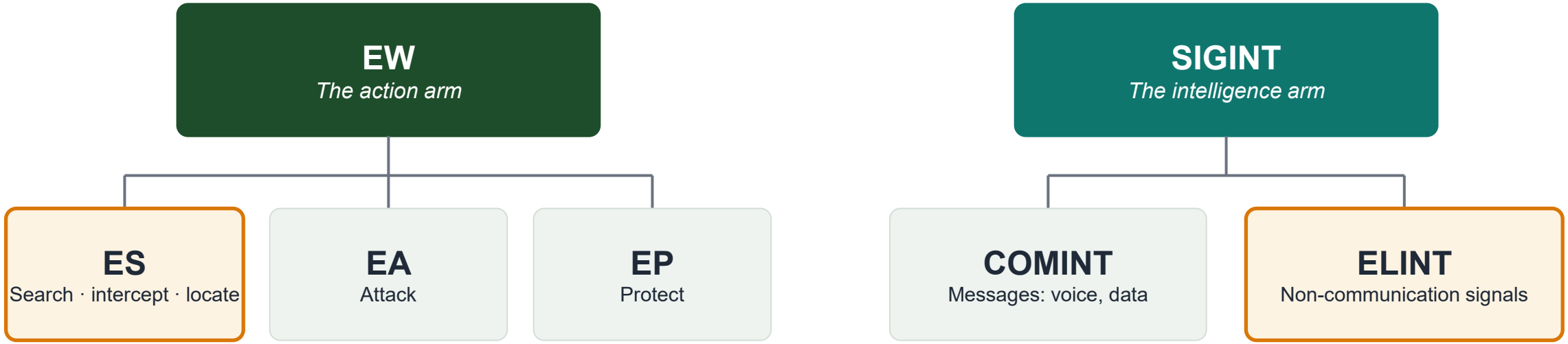

*Our DF bearings cross on an enemy radar*

- Every radar transmits to work. Every transmission can be found.
- Found radars become targets: Ukraine reports repeatedly destroying Russian Zoopark-1 counter-battery radars.
- The same is true of OUR radars.

**PAM 3-0702-01:** ELINT is “a specialized sub-discipline of SIGINT dedicated to the collection and technical analysis of non-communication electromagnetic signals.”

***A radar must shout to see. ELINT listens to the shout.***

# Where ELINT Sits




**ES and ELINT often use the same receivers.** The difference is who tasks them and why: ES for the commander's immediate action, ELINT for intelligence.

*PAM 3-0702-01 paras 0104–0105, 0118–0119*

# ELINT vs COMINT

## COMINT

*Communications Intelligence*

| | |
|---|---|
| Listens to | **People** |
| Signals | Voice, text, data messages |
| Skills | Language, cryptography |
| Answers | What does the enemy intend? |

## ELINT

*Electronic Intelligence*

| | |
|---|---|
| Listens to | **Machines** |
| Signals | Radars, beacons, sensors |
| Skills | Electronics, signal analysis |
| Answers | What equipment, where, doing what? |

**COMINT listens to the operators. ELINT listens to the hardware.**

# ES vs ELINT: Who Owns It?

## ELECTRONIC SUPPORT

*The backup camera*

- Owned by the tactical commander
- Real time, perishable
- Threat warning and immediate targeting

## ELINT (SIGINT)

*The black-box recorder*

- Owned by intelligence channels
- Deliberate, deep analysis
- Long-term picture of enemy capabilities

**Same receiver, different "why." Ask: who tasked it, and what is the product for?**

# What ELINT Delivers: The Electronic Order of Battle

**ES and ELINT feed the EOB: the catalogue of every emitter in the area (PAM 3-0702-01, para 0127).**

## FRIENDLY EOB

Our radios, radars, data links and navigation aids.

*Use: prevent mutual interference and protect OPSEC.*

## ADVERSARY EOB

Location, frequency and characteristics of hostile radios, weapon-guidance radars and EW systems.

*Use: drive targeting and plan countermeasures.*

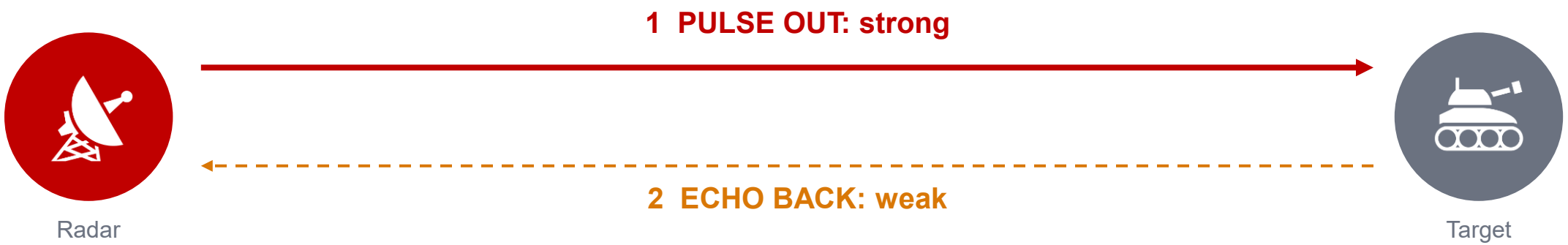
## NEUTRAL EOB

Civilian and commercial emitters: TV, air traffic control, maritime distress.

*Use: avoid hitting civilians; stay within telecom agreements.*

**ELINT's main job on the ground: fill and update the Adversary EOB.**

# How Radar Works in 90 Seconds




**3 TIME = RANGE**

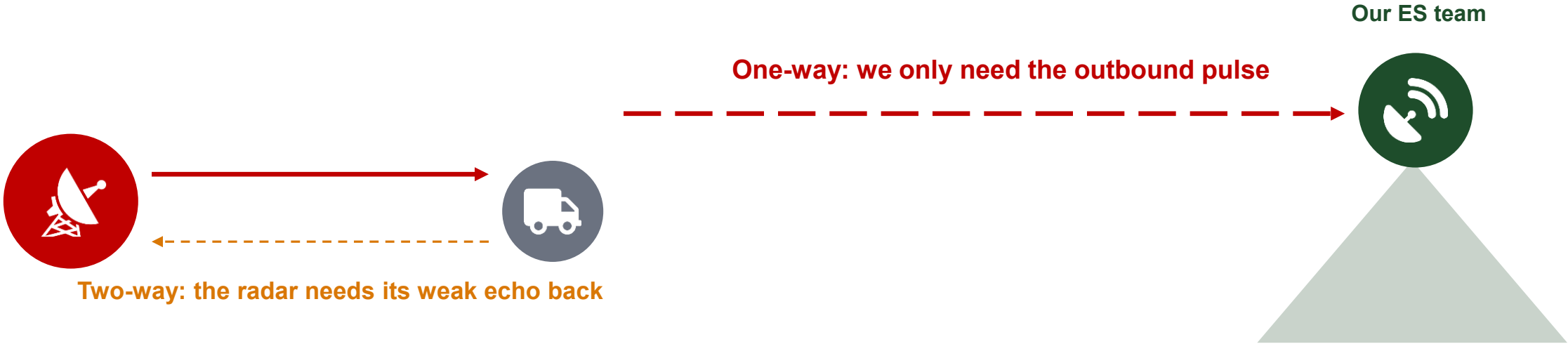
**Range = (speed of light × echo time) ÷ 2**

Echo after 100 μs → target about 15 km away

**Analogy:** shout into a canyon and time the echo. The longer the wait, the farther the wall.

**To see, a radar must transmit loudly. That loud pulse is what ELINT intercepts.**

# The One-Way Advantage




- Radar energy weakens on the way out AND on the way back.
- Our receiver needs only the outbound trip.
- So we can usually detect a radar from beyond the range at which it can detect us.

**Passive ≠ invisible.** Our own radios, vehicles and lights can still give our site away.

***You hear a sentry's footsteps long before his flashlight finds you.***

# The Radar Fingerprint

**Frequency (Ft)**
Which band it lives in

**Pulse width (PW)**
How long each pulse lasts

**PRI / PRF**
How often it pulses

**Modulation and bandwidth**
Shape of the signal inside the pulse

TRANSMITTER



*Each radar type has a recognisable set of parameters.*

**Scan type and period**
How the antenna moves

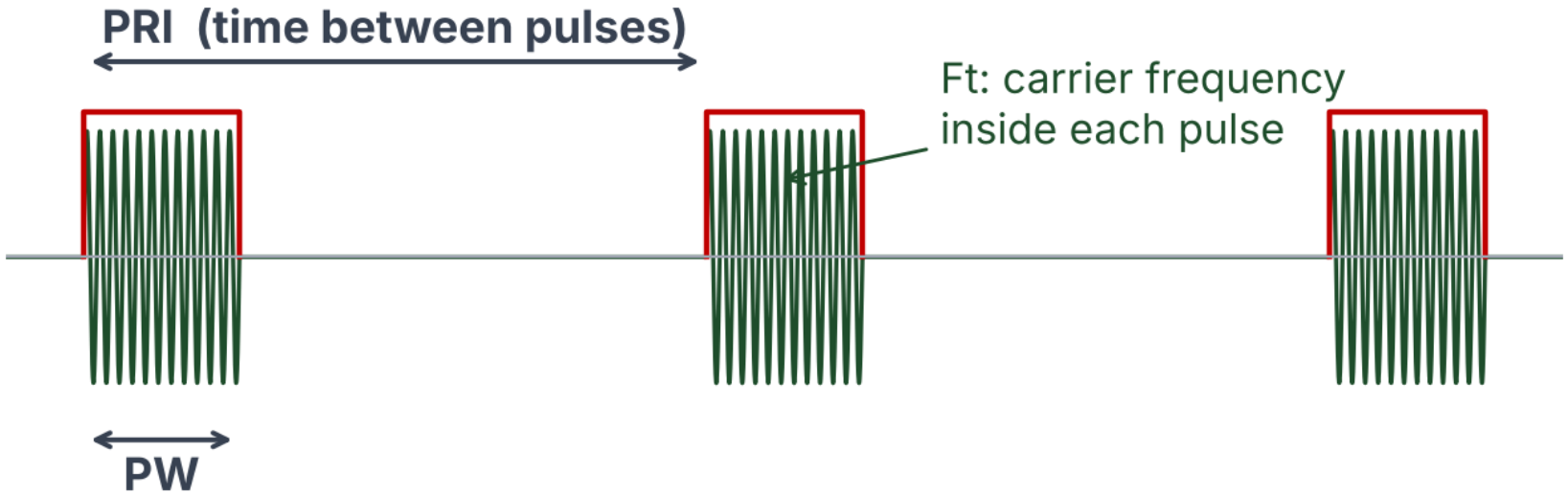
**Beamwidth**
How narrow the beam is

**Polarization**
Orientation of the wave

ANTENNA

*PAM 3-0702-01 para 0205 · AFPM 6-06 Ch 1*

# Reading the Pulse: Ft, PW, PRI




## Ft: Carrier frequency

Measured in MHz or GHz. Tells you the band and the radar family.

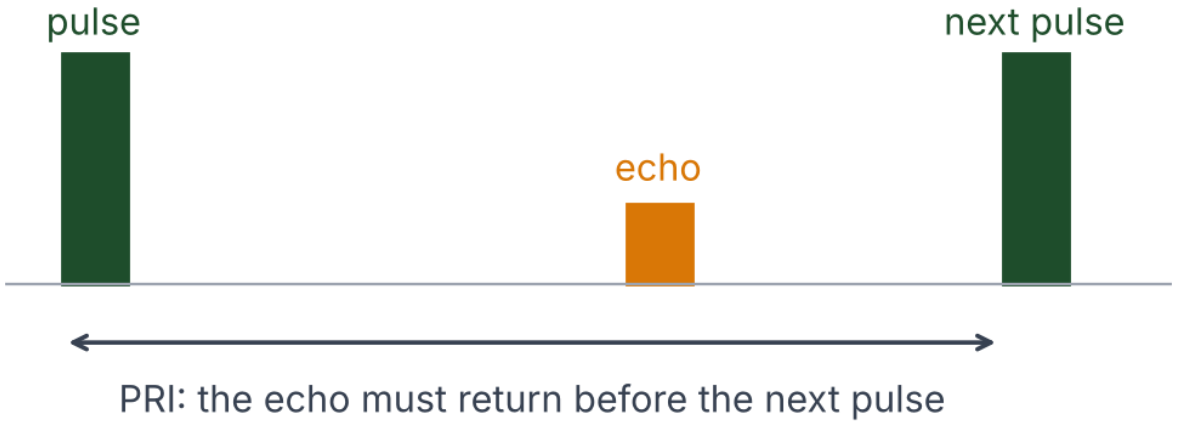
## PW: Pulse width

How long each pulse lasts, in microseconds (μs).

## PRI / PRF

PRI = time between pulses. PRF = pulses per second = 1 ÷ PRI.

# What the Pulse Tells You




Long PRI: the radar can see farther without confusing echoes (maximum unambiguous range).

Short PW: sharper range detail and a smaller blind zone close in.

Long PW: more energy on target, but a bigger blind zone close in.

**Average power = Peak power × PW ÷ PRI**
100 kW peak × 1 µs ÷ 1 ms = 100 W average

**Rule of thumb**
PRI 1 ms → about 150 km maximum unambiguous range

**The numbers reveal the job: long-range search or short-range precision.**

# Reading the Antenna: Scan and Beam

## Circular



**Circular:** 360° search. Sweeps the whole area like a lighthouse.

## Sector



**Sector:** Back and forth over an area of interest.

## Track (steady)



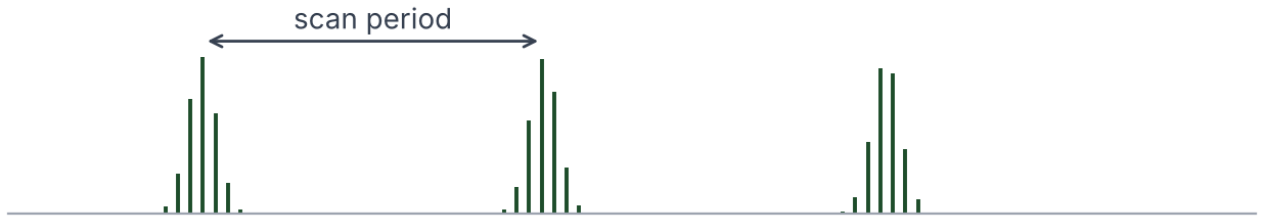
**Track (steady):** Beam held on one target. Engagement may follow.

**Beamwidth: a narrow beam gives precise direction; a wide beam covers more area faster.**

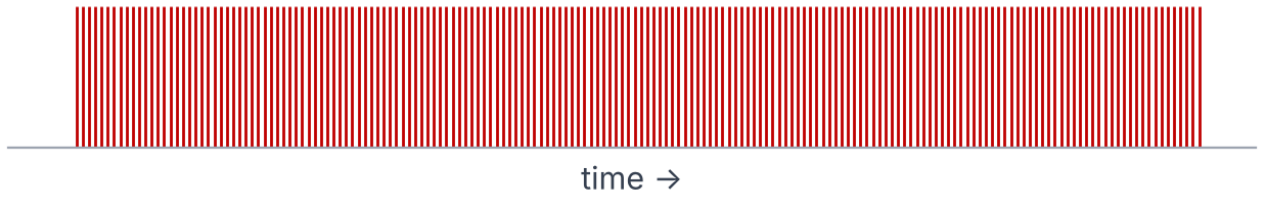
# Search or Strike? Reading Intent

## WHAT OUR RECEIVER SEES

**SEARCH: beam sweeps past you once per rotation**




**TRACK: beam stays on the target, pulses arrive non-stop, faster PRF**

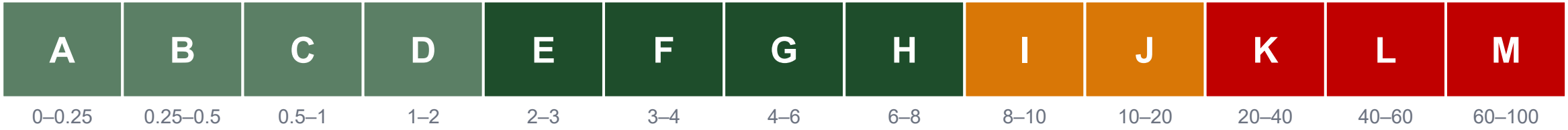



| | SEARCH | TRACK |
|---|---|---|
| Scan | Rotating or sector | Steady on target |
| PRF | Lower | Higher |
| PW | Longer | Shorter |
| Meaning | Looking around | Aiming at someone |

**Same emitter switches from SEARCH to TRACK: report immediately.**

***A lighthouse glances around. A sniper's scope stares at you.***

# Radar Bands: What Lives Where




*NATO bands, frequency in GHz*

**Long-range search and early warning**

**Air surveillance, air defence, weapon-locating**

**Tracking, fire control, ground and coastal surveillance**

**Short-range, high-precision sensors and seekers**

**Range depends on power, antenna and target, not on the band letter. These are typical uses only.**

**Lower frequency: sees farther, less precise. Higher frequency: more precise, shorter range.**

# Ground Emitters You Will Meet



## Weapon-locating radar

Detects incoming shells and computes where the gun fired from.

*Hunts our artillery. High-value target.*



## Ground-surveillance radar

Detects people and vehicles moving at range.

*Watches our routes and approaches.*



## Coastal surveillance radar

Tracks vessels along the coast.

*Common on our shores: friendly, civil or foreign?*



## Short-range air-defence radar

Searches and tracks low-flying aircraft and drones.

*Threat to our helicopters and drones.*



## Drone sensors

Radar altimeters and small radars on UAS.

*A radar signal can betray a drone nearby.*

**For each one: what is it, whose is it, and what is it doing right now?**

# The ES and ELINT Process

| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| **Search and Detect** | **Locate and Track** | **Analyse** | **Evaluate** | **Collate and Report** |
| Scan the band until a signal appears. | Take bearings from several sites and plot a fix. | Measure the fingerprint: Ft, PW, PRI, scan. | Match against the signature library. | Record, report, update the EOB. |

***Find it, follow it, fingerprint it, match it, file it.***

*PAM 3-0702-01 paras 0204–0205 · AFPM 6-06 Ch 1*

# Building the Signature Library

### BUILD
Every intercept becomes a record with time, parameters and location.

### VALIDATE
Check it isn't contamination from our own sensor or a known neutral emitter.

### SHARE
Feed the EOB and the EW reprogramming loop so jammers and warning receivers stay current.

**SIGNATURE RECORD (example, fictional)**

| | |
|---|---|
| Date / time | DD HHMM H MMM YY |
| Frequency | 9,410 MHz (I-band) |
| Pulse width | 0.8 µs |
| PRI / PRF | 1,000 µs / 1 kHz |
| Scan | Circular, 2.4 s |
| Bearing / fix | 285° from site; fix grid |
| Assessed system | Coastal surveillance radar |
| Confidence | Medium |

***Our "mugshot book" of enemy radars: only as good as each record we add.***

# Scenario: Coastal ES Team (Fictional)




*Northern Luzon coast. Schematic.*

**INTERCEPT LOG**

**0300H  New emitter**
Bearing 290°
9.41 GHz
PW 0.5 µs · PRF 1.2 kHz
Circular scan, 2.4 s

**0312H  Change**
Scan stops: steady signal
PRF rises to 4 kHz

1. **Which band is it in?**
2. **What changed at 0312H, and what might it mean?**
3. **What do you report, to whom, and how fast?**

**Class exercise: 3 minutes in pairs, then report back.**

# From Intercept to Effect

**ES / ELINT**

Detect, locate, fingerprint

**Intelligence**

Confirm identity, update EOB

**Commander**

Decide: targeting board

**Effect**

EA, fires, or EP measures

CONCEPT EXPLAINED WITH AN AIR ASSET

**SEAD:** aircraft use ELINT to locate air-defence radars, then jam or strike them so friendly aircraft can operate.

**Don't jam what you collect.** Frequencies being exploited are GUARDED; jamming them costs intelligence.

**PAM 3-0702-01: ES data feeds the targeting board to match emitters with EA, cyber or kinetic responses.**

# Key Takeaways

**1**

## Radars announce themselves

A radar must transmit to see, and we hear it from farther than it sees us.

**2**

## Parameters reveal identity and intent

Ft, PW, PRI and scan tell us what it is and whether it is searching or tracking.

**3**

## ELINT feeds the EOB

and the commander's next move, under PAM 3-0702-01.

***COMINT listens to the operators. ELINT listens to the hardware.***

# Check on Learning

1. Using PAM 3-0702-01, what are the two sub-disciplines of SIGINT?
2. What is the difference between ES and ELINT?
3. If the PRI is 1 ms, what is the PRF, and roughly how far can the radar see unambiguously?
4. Name two signs that a radar has switched from search to track.
5. What are the three Electronic Orders of Battle in PAM 3-0702-01?

# END OF PRESENTATION

*Electronic Intelligence (ELINT)*

## Questions?